


Exo 3. Q3. $\int_0^1 \frac{x^2}{\sqrt{x(1-x)}} dx$ $u = \sqrt{1-x}$ $du = -\frac{1}{2\sqrt{1-x}} dx$

$= 2 \int_0^1 (1-u^2)^{3/2} du$ $\uparrow$ $i.e. x = 1-u^2$ $2 \int_0^{\pi/2} (\sin^2 t)^{3/2} \sin t dt$

$u = \cos t$ $du = -\sin t dt$

$\sin^4(t) = \left(\frac{e^{it} - e^{-it}}{2i} \right)^4 = \frac{e^{i4t} + e^{-i4t} - 4e^{i3t}e^{-it} - 4e^{-i3t}e^{it} + C_4 e^{i2t}e^{-i2t}}{16}$

$= \frac{\cos(4t)}{8} - \frac{4}{16} (e^{2it} + e^{-2it}) + \frac{6}{16}$

$= \frac{\cos(4t)}{8} - \frac{1}{2} \cos 2t + \frac{3}{8}$

$\int_0^{\pi/2} \cos(4t) dt = \left[\frac{\sin 4t}{4} \right]_0^{\pi/2} = 0$ $\text{idem} \int_0^{\pi/2} \cos 2t dt = 0.$

Donc $\frac{2}{\pi} \int_0^{\pi/2} \sin^4 t dt = \frac{2}{\pi} \times \frac{\pi}{2} \times \frac{3}{8} = \frac{3}{8}.$

4. Calculer la CDF F de X . Pour $x \in (0,1)$.

$F(x) = \frac{1}{\pi} \int_0^x \frac{1}{\sqrt{t(1-t)}} dt$ $\uparrow$ $\frac{1}{\pi} \int_0^{\sqrt{x}} \frac{2}{\sqrt{1-t^2}} dt = \left[\frac{2}{\pi} \arcsin t \right]_0^{\sqrt{x}}$

$u = \sqrt{t}$ $= \frac{2}{\pi} \arcsin \sqrt{x}$

$q_{1/4} = \frac{2 - \sqrt{2}}{4}$

$q_{1/2} = \frac{1}{2}$ car $P(X < \frac{1}{2}) = P(1-X < \frac{1}{2}) = P(X > \frac{1}{2})$.

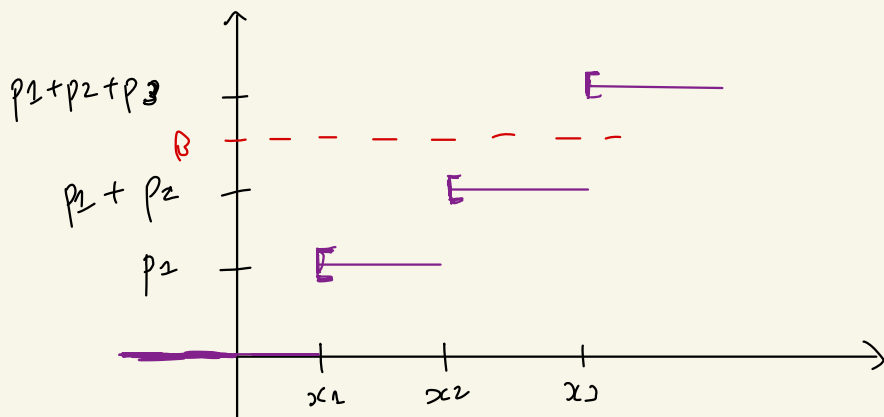
$x = \frac{1}{2}.$

$$P(X < q_{1/4}) = \frac{1}{4}$$

$$\text{i.e. } P(1-X < q_{1/4}) = \frac{1}{4} \quad \text{i.e. } P(X > 1 - q_{1/4}) = \frac{1}{4}$$

$$\text{i.e. } P(X < 1 - q_{1/4}) = \frac{3}{4} \Rightarrow q_{3/4} = 1 - q_{1/4}.$$

Exo 4. On suppose que $x_1 < x_2 < \dots < x_n < \dots$ et $p_i > 0 \forall i$.



Soit $\beta \in (0, 1)$. Si $\beta \leq p_1$ alors $F^{(-1)}(\beta) = x_1$.

détails: $F(x) = 0$ pour $x < x_1$

$$\text{et } F(x_2) = p_1 \geq \beta$$

Donc $F^{(-1)}(\beta) = \inf \{x \in (0, 1) : F(x) \geq \beta\} = x_1$.

• Sinon $\exists i \geq 1$ tq. $\sum_{k=1}^i p_k < \beta \leq \sum_{k=1}^{i+1} p_k$. (On a supposé $p_i > 0 \forall i$)

$$\text{i.e. } F(x_i) < \beta \leq F(x_{i+1})$$

On a aussi $F(x) = F(x_i) \quad \forall x \in [x_i, x_{i+1}[$.

Donc $F^{(-1)}(\beta) = \inf \{x \in (0, 1) : F(x) \geq \beta\} = x_{i+1}$.

2) $X = \sum_{i=1}^n X_i$ $X_i \stackrel{i.i.d}{\sim} \text{Be}(p, 5)$ X_i symétrique 7/2 à 1/2:

Donc $\sum_{i=1}^n (X_i - \frac{1}{2}) \sim \sum_{i=1}^n (\frac{1}{2} - X_i)$

i.e. $X - \frac{n}{2} \sim \frac{n}{2} - X$ loi sym 7/2 à 1/2.

Si $n=10$, $\frac{n}{2} = 5$ est la médiane:

$$P(X - 5 < 0) = P(X < 5) = P(5 - X < 0) \\ = P(X > 5).$$

De m $P(X \leq 5) = P(X \geq 5)$.

De $P(X \leq 5) > \frac{1}{2}$ et $P(X < 5) < \frac{1}{2}$. (donc $F(5) > \frac{1}{2}$ et $F(x) < \frac{1}{2}$ $\forall x < 5$)

(En effet: $P(X \leq 5) = 1 - P(X > 5)$
et $P(X > 5) + P(X < 5) + P(X = 5) = 1$

i.e. $2P(X > 5) = 1 - P(X = 5)$

De $P(X > 5) < \frac{1}{2}$.)

m même pour tout n pair de $\frac{n}{2}$ et une médiane de $\text{Bin}(n, 1/2)$.